REPORT LAB 9:

MEASURING FOCAL LENGTHS OF CONVEX AND CONCAVE LENSES

Class: CC10 / Group: 01	Lecturer's comment
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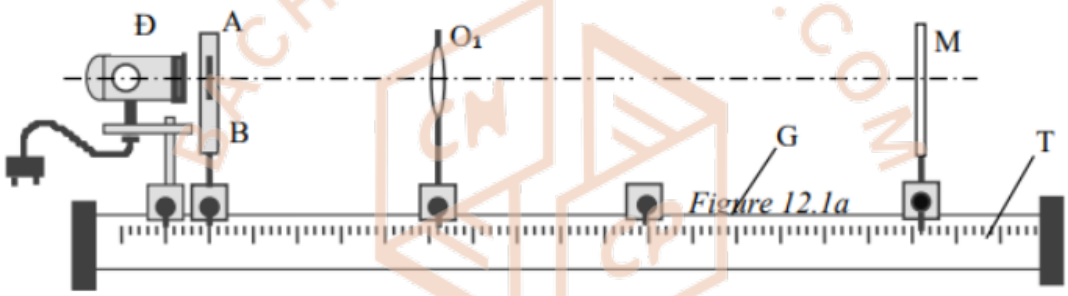


Figure .1

I. Aims/Purposes

1. Look through the convergent lens at the picture that is developing. Measure the focal length of the lens.

2. Examine how an image is formed using a lens system that includes a convergent lens and a divergent lens.

3. Measure the focal length of the divergent lens.

II. Apparatus, Methods, and Procedure

1. Apparatus:

- 1 optical bench 1000mm
- 1 convergent len O1
- 1 divergent len O2
- 1 light source Đ type 6V-8W
- 1 power source 6V-3A
- 1 AB object shape like the number "1" inside a circular hole of the plastic plate (H.1).
- 1 screen M 70x100mm.

2. Procedure:

➢ *Measuring the focal length of convergent lens*:

❖ **First method: Silberman method:**

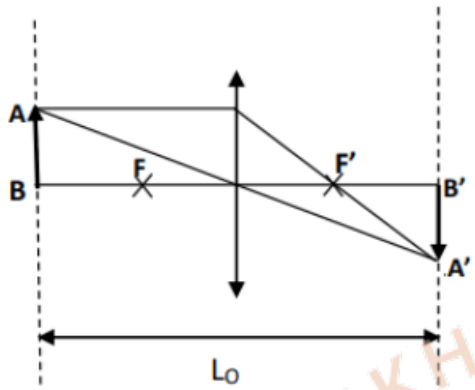


Figure .2

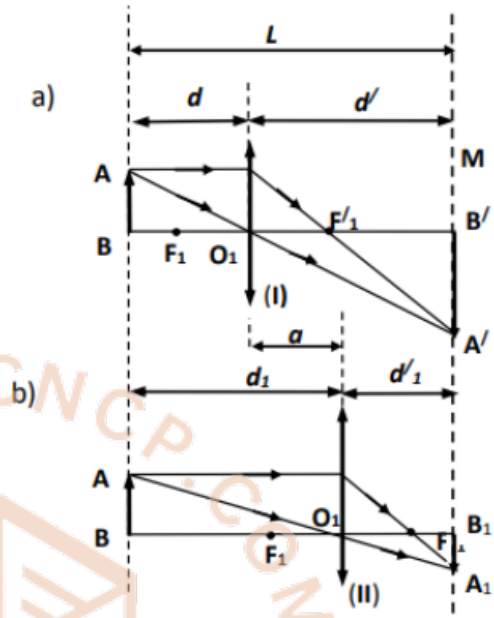


Figure. 3

a) At the 10cm mark, place AB near the light. After that, position the AB object to obtain a uniform exposure. The convergent lens is situated between the AB and the screen M with a mutual distance of less than 4f.

b) Adjust the convergent lens O1 and the screen M so that the lens O1 and the AB and M are at the same distances until the picture is crisp. The picture is the same size as the object in this setup. (Moving the lens for a centimeter and the screen doubles the lens' moving distance till it's near the sharpest condition, then moving with a millimeter-scale to get the exact location.) To table 1 in the report, write down the distance L0 between AB and the screen M.

c) Make replications of the measurement in (b) triple times. In this case, focal length f_1 of the convergent lens O_1 can be determined via:

$$f_{1_i} = \frac{L_{oi}}{4} \qquad (12.3)$$

❖ **Second method: Besse lmethod:**

a) Place the screen M far away AB with distance L=4. f on the optical bench (in this experiment, the suggested values are L= $4{,}5f_1$, L= $4{,}7f_1$, L= $4{,}9f_1$)

b) Moving the convergent lens O_1 from near AB object to farther away to position (I) at which the sharpest image $A^{/}B^{/}$ is larger than the AB object on the screen M (Figure.3a). Write down the coordinate x_1 of the lens O_1 at position (I) in table 1 of the report.

c) Continue moving the convergent lens O_1 farther away from AB to position (II) to obtain the sharpest image A_1B_1 smaller than AB on the screen M (Figure 3b). Write down the coordinate x_2 of the lens O_1 at the position in) to table 1 of the report.

d) Make replication of the task (b) and (c) triple times. In the above, the displacement a of the lens O_1 from position (I) to position (II) is:

$$a = x_2 - x_1$$

and the focal length f_1 of the convergent lens O_1 can be determined by:

$$f_{1_i} = \frac{L_i^2 - a_i^2}{4L_i}$$

➢ ***Measuring the focal length of the divergent lens: the connecting point method***

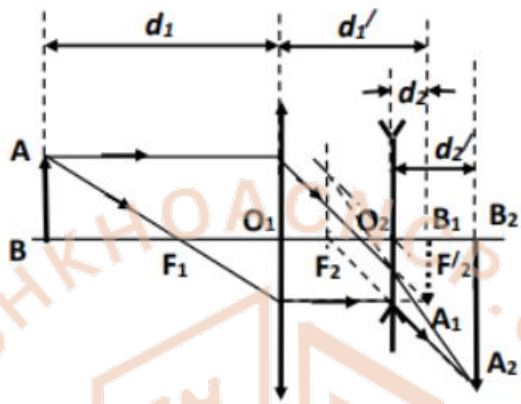


Figure .4

Divergent lenses only provide the real image of the imaginary object. Therefore, to measure the focal length f_2 of the divergent lens O_2, we need to couple it with another convergent lens O_1 to be a coaxial lens system so that the real image A_1B_1 of the AB object given by the convergent lens O_1 located back within the focal length $|f_2|$ of the divergent lens O_2 (Figure. 4) so the image A_1B_1 become the imagined object of the lens O_2 by follows:

a) Keeping the position of the AB object and the convergent lens O_1 at the position (tor to obtain the sharpest image A_1B_1 smaller than AB on the screen M in Figure 12.3b. Place the divergent lens O_2 on track 4 at the back of the convergent lens O_1 and being coaxial with lens O_1, apart from the screen M distance $|d_2| = O_2B_1 < |f_2|$ (in this experiment, the suggested values are $|d_2|$ = 50mm,55mm,60mmm).

b) Moving the screen M farther away from the dive from urgent lens O_2 to the position M/ to obtain the sharp image A_2B_2 apart from divergent lens O_2 distance d'_2 in Figure 12.4. Make replication of this task triple times and write down the distance d'_2 every time with the corresponding value of d_2 in table 12.2 of the report.

- FocalThe focal length f_2 of the divergent lens O_2 can be determined by:

$$f_2 = \frac{d_2 . d_2'}{d_2 + d_2'}$$

III. Equations

✓ **Silberman's method:**

$$f_{1_i} = \frac{L_{oi}}{4}$$

$\Delta f_{1\,sys} = \frac{\Delta L_{o\,sys}}{4}$ (mm)

$\Delta f_1 = \overline{\Delta f_1} + \Delta f_{1\,sys}$ (mm)

✓ **Bessel's method:**

$f_{1_i} = \frac{L_i^2 - a_i^2}{4L_i}$ (mm)

$\frac{\Delta f_{1\,sys}}{f_{1\,mid}} = \left|\frac{2L_{mid}}{L_{mid}^2 - a_{mid}^2} - \frac{1}{L_{mid}}\right| \Delta L_{sys} + \left|\frac{-2a_{mid}}{L_{mid}^2 - a_{mid}^2}\right| \Delta a_{sys}$ (mm)

$\Delta f_1 = \overline{\Delta f_1} + \Delta f_{1\,sys}$ (mm)

✓ **Connecting point method:**

$f_{2_i} = \frac{d_{2_i} . d'_{2_i}}{d_{2_i} + d'_{2_i}}$ (mm)

$\frac{\Delta f_{2\,sys}}{|f_{2\,mid}|} = \left|\frac{1}{d_{2\,mid}} - \frac{1}{d_{2\,mid} + d'_{2\,mid}}\right| \Delta d_{2\,mid} + \left|\frac{1}{d'_{2\,mid}} - \frac{1}{d_{2\,mid} + d'_{2\,mid}}\right| \Delta d'_{2\,mid}$ (mm)

$\Delta f_2 = \overline{\Delta f_2} + \Delta f_{2\,sys}$ (mm)

IV. Experimental data

Table 1

	Silberman method			Bessel method			
	L_0(mm)	**f_1(mm)**	**Δf_1(mm)**	**L(mm)**	**a(mm)**	**f_1(mm)**	**Δf_1(mm)**
1	400	100.0	0.33	450	150	100.0	0.17
2	398	99.5	0.17	470	179	100.5	0.33
3	398	99.5	0.17	490	210	100.0	0.17
Average	398.7	99.67	0.22			100.17	0.22

Table 2

	Connecting point method			
	d_2(mm)	**d'_2(mm)**	**f_2(mm)**	**Δf_2(mm)**
1	-50	175	-70.0	0.13
2	-55	258	-69.9	0.03
3	-60	430	-69.7	0.17
Average			-69.87	0.11

V. Calculation works

Focal length and its uncertainties

Silberman method:

$$f_{11} = \frac{L_{01}}{4} = \frac{400}{4} = 100.0\ (mm)$$

$$f_{12} = \frac{L_{02}}{4} = \frac{398}{4} = 99.5\ (mm)$$

$$f_{13} = \frac{L_{03}}{4} = \frac{398}{4} = 99.5\ (mm)$$

$$\overline{f_1} = \frac{f_{11} + f_{12} + f_{13}}{3} = \frac{100.0 + 99.5 + 99.5}{3} \approx 99.67\ (mm)$$

$$\Delta f_{1sys} = \frac{\Delta L_{sys}}{4} = \frac{1}{4} = 0.25\ (mm)$$

$$\Delta f_1 = \Delta f_{1sys} + \overline{\Delta f_1} = 0.25 + 0.22 = 0.47\ (mm)$$

Bessel method:

$$f_{11} = \frac{{L_1}^2 - {a_1}^2}{4L_1} = \frac{450^2 - 150^2}{4 \times 450} = 100.0\ (mm)$$

$$f_{12} = \frac{{L_2}^2 - {a_2}^2}{4L_2} = \frac{470^2 - 179^2}{4 \times 470} = 100.5\ (mm)$$

$$f_{13} = \frac{{L_3}^2 - {a_3}^2}{4L_3} = \frac{490^2 - 210^2}{4 \times 490} = 100.0\ (mm)$$

$$\overline{f_1} = \frac{f_{11} + f_{12} + f_{13}}{3} = \frac{100.0 + 100.5 + 100.0}{3} \approx 100.17\ (mm)$$

$$\frac{\Delta f_{1sys}}{f_{1mid}} = \left|\frac{2L_{mid}}{L_{mid}^2 - a_{mid}^2} - \frac{1}{L_{mid}}\right| \Delta L_{sys} + \left|\frac{-2a_{mid}}{L_{mid}^2 - a_{mid}^2}\right| \Delta a_{sys}$$

$$= \left|\frac{2 \times 470}{470^2 - 179^2} - \frac{1}{470}\right| \times 1 + \left|\frac{-2 \times 179}{470^2 - 179^2}\right| \times 1 \approx 4.75\text{x}10^{-3}$$

$$\Rightarrow \Delta f_{1sys} = 4.75\text{x}10^{-3} \times f_{1mid} = 4.75\text{x}10^{-3} \times 100.5 \approx 0.48\ (mm)$$

$$\Delta f_1 = \Delta f_{1sys} + \overline{\Delta f_1} = 0.48 + 0.22 = 0.70\ (mm)$$

Connecting points method:

$$f_{21} = \frac{d_{21} d'_{21}}{d_{21} + d'_{21}} = \frac{-50 \times 175}{-50 + 175} = -70.0\ (mm)$$

$$f_{22} = \frac{d_{22} d'_{22}}{d_{22} + d'_{22}} = \frac{-55 \times 258}{-55 + 258} \approx -69.9\ (mm)$$

$$f_{23} = \frac{d_{23}d'_{23}}{d_{23} + d'_{23}} = \frac{-60 \times 430}{-60 + 430} \approx -69.7\ (mm)$$

$$\overline{f_2} = \frac{f_{21} + f_{22} + f_{23}}{3} = \frac{-70.0 - 69.9 - 69.7}{3} \approx -69.87\ (mm)$$

$$\frac{\Delta f_{2sys}}{|f_{2mid}|} = \left|\frac{1}{d_{2mid}} - \frac{1}{d_{2mid} + d'_{2mid}}\right| \Delta d_{2sys} + \left|\frac{1}{d'_{2mid}} - \frac{1}{d_{2mid} + d'_{2mid}}\right| \Delta d'_{2sys}$$

$$= \left|\frac{1}{-55} - \frac{1}{-55 + 258}\right| \times 1 + \left|\frac{1}{258} - \frac{1}{-55 + 258}\right| \times 1 \approx 2.42\text{x}10^{-2}$$

=> $\Delta f_{2sys} = 2.42\text{x}10^{-2} \times 69.9 \approx 1.69\ (mm)$

$\Delta f_2 = \Delta f_{2sys} + \overline{\Delta f_2} = 1.69 + 0.11 = 1.80\ (mm)$

VI. Results

Silberman method: $f_1 = \overline{f_1} \pm \Delta f_1 = 99.67 \pm 0.47\ (mm)$
Bessel method: $f_1 = \overline{f_1} \pm \Delta f_1 = 100.17 \pm 0.70\ (mm)$
Connecting points method: $f_2 = \overline{f_2} \pm \Delta f_2 = -69.87 \pm 1.80\ (mm)$

VII. Question

1. Write down convex lens formulas and sign consent of all quantities in these formulas.

- Lens Formula:

$$\boxed{\frac{1}{f} = \frac{1}{d} + \frac{1}{d'}}$$

- $f > 0$: *converging lens*
- $f < 0$: *diverging lens*
- $d > 0$: *real object* | $d < 0$: *imaginary object*
- $d' > 0$: *real image* |$d' < 0$: *imaginary image*

$$k = \frac{d'}{d}$$

- $d > 0$ and $d' > 0$: *real object and real image*
  - ⇨ $k < 0$: *object and image inverted*
- $d < 0$ and $d' < 0$: *imaginary object and imaginary image*
  - ⇨ $k > 0$: *object and image are in the same direction*

2. Describe the measurement method of the convergent lens in this experiment. Draw figure illustrating the image forming of the AB object.

❖ **First method: Silberman method:**

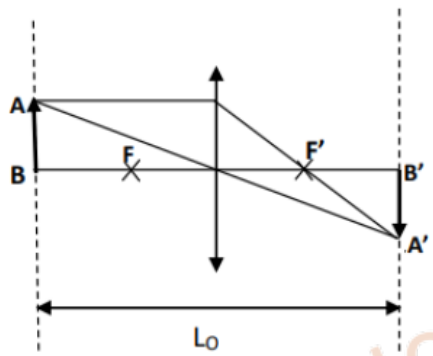
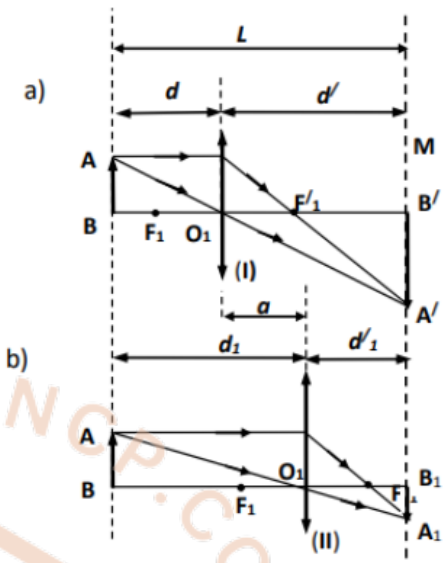


Figure .2



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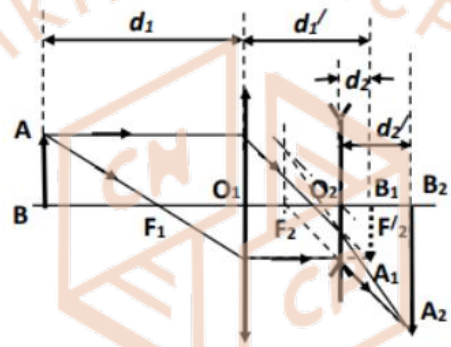

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- Moving the screen M farther away from the dive from urgent lens O_2 to the position $M^/$ to obtain the sharp image A_2B_2 apart from divergent lens O_2 distance $d^/_2$ in Figure 12.4. Make replication of this task triple times and write down the distance $d^/_2$ every time with the corresponding value of d_2 in table 12.2 of the report.
- The focal length f_2 of the divergent lens O_2 can be determined by:

$$f_2 = \frac{d_2 . d_2'}{d_2 + d_2'}$$

4. Prove that if the distance L between AB and its real image given by the convergent lens with focal length f1 is constant, :
- if L = 4f₁ : there is only one position of convergent lens within L providing the sharpest image on the screen M.
- if L > 4f₁ : there are two positions of convergent lens within distance L that provide sharp image.

- We have: $L = |d + d'| = d + d'$ because we consider the case of a lens for an image on a screen (real image)

=> $L = d + \frac{df}{d-f} = \frac{d^2}{d-f}$

=> $d^2 - Ld + Lf = 0$

=> $\Delta = L^2 - 4Lf$

- There are two positions of the lens to produce a sharp image on the screen, ie the above quadratic equation must have 2 distinct solutions of d. The condition for that is **$\Delta > 0 \rightarrow L > 4f$.**

- According to Viet's theorem, we have 2 solutions d1, d2 with a sum of :

$$d_1 + d_2 = -\frac{b}{a} = D$$

- On the other hand, we have:

$$d_2 - d_1 = 1 => d_2 = \frac{D+1}{2} \; and \; d_2 = \frac{D-1}{2}$$

$$=> \; d_1' = D - d_1 = \frac{D+1}{2} => f = \frac{(D-1)(D+1)}{4D} = \frac{D^2 - 1^2}{4D}$$

- So, to get two positions of the lens for a clear image on the screen, the condition is that the object - screen distance must be greater than 4f.

- Especially if $D = 4f$ $(when\ \Delta = 0)$ then there is only one position of the lens for a clear image on screen E. Position I places the lens as the midpoint of the object-screen distance. The two positions of the lens provide a clear image on the screen that is symmetrical about the center point I.

+ The object-image permutation at two positions of the lens: $d_1 = d_2'$ and $d_2 = d_1'$

+ Magnification at two positions of the lens: $k_1\ k_2 = 1$ and the height of the object: $AB = \sqrt{A_1B_1A_2A_2}$.

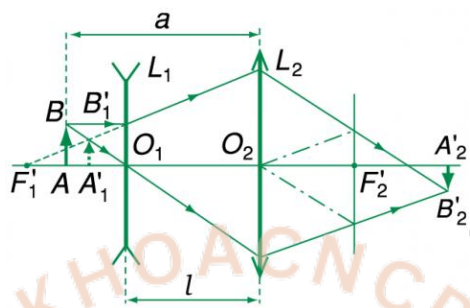
⇨ Summary:

- ❖ If $\Delta = 0$ => L = 4f1: there is only **one position** of the convergent lens **(one value of d)** within L providing the sharpest image on the screen M.
- ❖ If $\Delta > 0$ => L > 4f1: there are two positions of the convergent lens **(two value of d)** within distance L that provide sharp image.

6. Can we measure the focal length of the divergent lens by placing this lens in front of the convergent lens (given all lenses are coaxial)? Draw illustrating figure for the image formation of this case.

- The focal length of the converging lens L_0 can be determined when performing this experiment.
- After obtaining the real image A'B' larger than the object AB clearly visible on the screen M, we must measure the distances d0 from the object AB and the distance d'0 from the real image A'B' to the converging lens L0. to calculate the focal length f0 of this lens using the formula.

$$f = \frac{dd'}{d + d'}$$



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